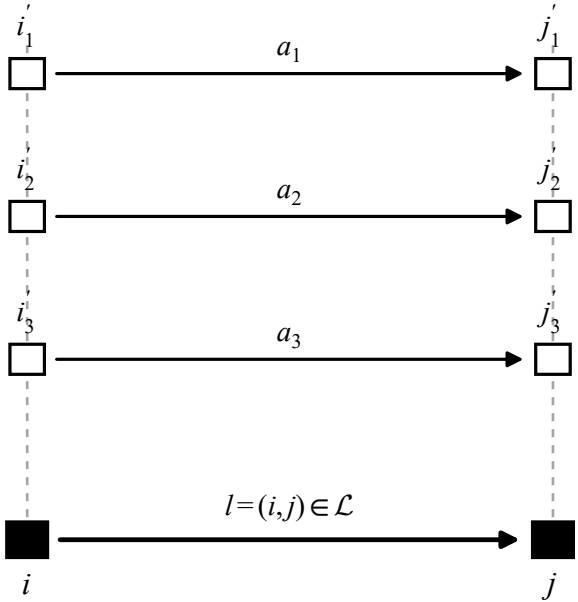
Service network



Physical network

$i'_k, j'_k \in \mathcal{V}$

$a_k \in \mathcal{A}$

$$s(i'_k)=i, \quad s(j'_k)=j, \quad \ell(a_k)=l \quad (k=1,2,3)$$